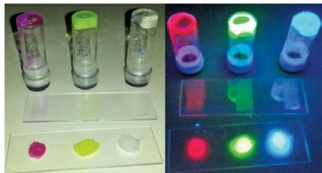


The researchers tested the sensors in baths of saline solution that caused these to dissolve after a few days. Next, they tested the devices in the brains of rats. Having seen that the sensors are accurate and dissolve in the solution and in the brains of rats, researchers now are planning to test the technology on humans.

Bio-inspired LEDs glow using proteins

A new type of hybrid light emitting diode (HLED) termed BioLED has been developed by a team of German-Spanish scientists. It features protein cascade coatings in the form of rubber to make the LEDs glow. Drawing inspiration from nature's bio-molecules, the scientists developed this hybrid



Rubber with red, green and blue luminescent proteins are used to produce the BioLEDs (Image courtesy: www.materialsgate.de)

device that introduces luminescent proteins into a polymer matrix to produce luminescent rubber. It uses luminescent proteins to convert the blue light emitted by a regular LED into pure white light. The rubber is used to cover the LED to convert the light.

Manufacturing of LEDs involves inorganic components such as cerium and yttrium that are already in short supply and hence would not be sustainable for long. BioLEDs are easy to manufacture, are bio-degradable and can be efficiently recycled and replaced. These also come with less than ten per cent loss in luminous efficiency for over 100 hours.

Lithium-ion battery that automatically shuts down before over-heating

A research group comprising chemical engineers from Stanford University has developed a lithium-ion battery that automatically shuts down before over-heating, and starts charging once the temperature falls.

Zhenan Bao, team leader at CE research group at Stanford, and postdoctoral scholar Zheng Chen, turned to nanotechnology to look for a reversible and long-

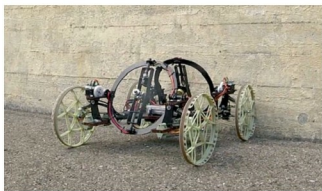
lasting solution. They developed a fast and reversible thermo-responsive polymer switching material that can be incorporated inside batteries to prevent thermal runaway. The material consists of conductive graphene-coated spiky nanostructure (nano-spiky) nickel particles as the conductive filler, along with a polymer matrix having a large thermal expansion coefficient. These nano-spikes have high electrical conductivity and high thermal sensitivity.

In order to conduct electricity, the nano-spikes must be in close vicinity. As temperature rises, the polythene stretches, causing the spikes to move apart from each other and thereby break electrical contact.

While experimenting, when the battery was heated up to 70°C, the polythene film quickly expanded like a balloon, shutting off the battery. But when the temperature dropped down to 70°C, the polyethylene shrunk, particles came in contact and the battery started regenerating electricity.

Disney's robot can climb walls

Disney Research Zurich, along with ETH, an engineering, science, technology, mathematics and management university, has developed a robot that can climb walls.



VertiGo, a wall-climbing robot including ground-wall transition (Image courtesy: www.disneyresearch.com)

It is called VertiGo and is capable of making a near-seamless transition from the ground to a wall. It is also capable of mind-boggling wall-riding owing to a pair of tiltable propellers that provide thrust onto the wall, effectively sucking it against the vertical surface.

With two steerable wheels, VertiGo can be operated as a radio-controlled car. But, unlike the radio-controlled car, VertiGo has two infrared distance sensors mounted on the front to estimate its orientation in space.

The robot has potential use in entertainment, such as providing visual effects, but is also a general technology for locomotion on walls with possible other uses such as industrial inspection.